

SYDNEY BADGERYS CREEK, Australia

WMO: 947520

Lat: 33.897S

Lon: 150.728E

Elev: 269

StdP: 14.55

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.8	37.1	27.9	22.0	58.1	30.0	24.3	55.0	24.6	62.0	21.2	60.1	2.8	230	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	96.1	69.8	91.5	69.0	87.2	68.2	74.4	86.2	72.8	83.4	71.5	81.4	10.7	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.6	117.7	76.8	70.0	111.6	75.7	68.7	106.5	74.7	38.0	86.8	36.6	83.3	35.4	81.3	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	17.5	15.1	30.5	105.5	1.4	4.6	29.5	108.8	28.6	111.5	27.8	114.0	26.8	117.4		
(5)			30.0	78.1	1.8	3.3	28.8	80.5	27.7	82.4	26.7	84.3	25.5	86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.8	73.5	72.8	69.4	63.8	57.1	52.9	51.3	53.4	58.7	63.2	67.5	70.9		
	DBStd	9.04	5.65	5.07	4.39	4.35	3.93	3.51	3.47	3.63	4.83	5.47	5.85	5.45		
	HDD50	46	0	0	0	0	2	12	23	9	0	0	0	0		
	HDD65	1831	3	2	10	72	247	363	424	361	200	101	38	10		
	CDD50	4723	728	639	602	413	221	99	64	113	260	410	525	649		
	CDD65	1033	266	221	147	35	1	0	0	0	10	47	112	194		
	CDH74	8610	2210	1620	876	276	22	0	2	12	217	636	1102	1637		
	CDH80	3388	1015	647	241	61	1	0	0	0	54	224	457	688		
(14) Wind	WSAvg	6.2	6.5	6.2	5.7	5.6	5.3	5.9	5.9	6.7	6.7	6.6	6.7	6.6		
(15) Precipitation	PrecAvg	31.3	3.7	3.5	3.3	2.8	2.4	2.3	1.3	1.9	1.7	2.4	3.1	2.4		
	PrecMax	48.9	10.6	15.9	12.2	14.4	7.3	12.3	5.2	9.5	5.7	7.2	14.6	7.3		
	PrecMin	14.6	0.4	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0		
	PrecStd	10.2	2.6	3.4	2.6	3.0	2.1	2.8	1.2	2.3	1.3	2.0	2.9	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.1	99.8	91.2	86.5	76.7	69.2	71.8	75.5	86.5	92.3	98.7	99.8		
		MCWB	71.9	71.6	70.1	65.2	60.8	56.4	55.0	56.4	60.8	62.8	65.8	69.4		
	2%	DB	95.9	92.9	85.6	80.6	73.0	65.8	66.5	70.7	80.1	86.1	91.0	93.1		
		MCWB	71.1	72.0	69.9	63.9	59.6	55.7	53.3	55.1	59.3	62.8	67.2	68.7		
	5%	DB	90.0	86.9	81.9	76.8	70.0	63.7	63.7	67.3	75.3	81.2	84.5	87.2		
		MCWB	70.5	70.8	69.1	63.7	58.5	54.4	52.3	53.6	58.4	62.3	66.0	67.1		
	10%	DB	85.1	82.9	78.6	73.7	67.5	61.7	61.3	64.3	71.0	76.1	80.0	82.9		
		MCWB	69.5	69.7	67.4	62.5	57.0	53.6	51.3	52.2	57.1	61.0	64.8	66.9		
	0.4%	WB	76.1	78.1	75.8	69.2	65.5	60.6	60.3	60.1	65.0	68.8	72.6	73.8		
		MCDB	92.0	91.6	81.9	79.6	69.7	63.8	64.8	68.9	74.7	80.5	85.6	87.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	74.0	74.5	72.8	67.1	62.5	58.3	56.7	57.5	62.6	66.3	70.1	71.9		
		MCDB	87.9	85.4	79.3	74.7	69.5	62.1	61.4	65.9	73.4	77.2	82.2	84.7		
	5%	WB	72.4	72.7	70.8	65.5	60.3	56.9	54.5	55.6	60.8	64.5	68.3	70.4		
		MCDB	84.5	81.9	78.3	72.5	67.5	61.0	60.3	63.5	70.4	74.9	79.6	81.7		
	10%	WB	71.0	71.3	69.2	64.2	58.6	55.5	52.8	53.9	59.0	62.8	66.6	68.9		
		MCDB	81.4	79.5	76.0	70.9	65.4	59.6	59.0	61.6	67.9	72.8	76.4	78.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	19.0	18.1	20.5	22.0	19.1	22.0	23.6	24.3	23.2	21.3	21.1		
		MCDBR	32.3	28.0	24.5	25.7	25.9	22.6	26.4	29.4	32.5	33.8	31.7	30.9		
		MCWBR	10.3	10.1	10.1	11.2	13.6	13.1	14.8	15.2	14.0	13.1	11.5	10.4		
	5% WB	MCDBR	25.4	22.8	20.2	21.1	21.9	17.5	21.0	24.6	26.8	25.9	25.4	24.9		
		MCWBR	9.3	9.6	9.4	10.3	12.6	11.5	13.3	13.9	13.0	11.6	10.5	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.394	0.390	0.361	0.332	0.315	0.305	0.296	0.303	0.335	0.357	0.383	0.394		
	taud		2.432	2.463	2.523	2.577	2.573	2.601	2.609	2.581	2.490	2.459	2.429	2.408		
	Ebn at Noon		299	294	291	283	271	266	276	289	295	301	301	300		
	Edh at Noon		39	36	32	27	24	22	23	26	32	36	39	40		
(44) All-Sky Solar Radiation	RadAvg		2033	1730	1483	1209	964	760	891	1176	1564	1807	1908	2090		
	RadStd		159	163	134	105	65	67	67	72	94	174	179	195		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		+0.79	N/A	N/A	N/A	N/A	-4	-153	+331	+176

Nomenclature: See separate page